Q:4 Image Sensor

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Define image sensor. With neat sketches, describe image acquisition by different arrangements of sensors.
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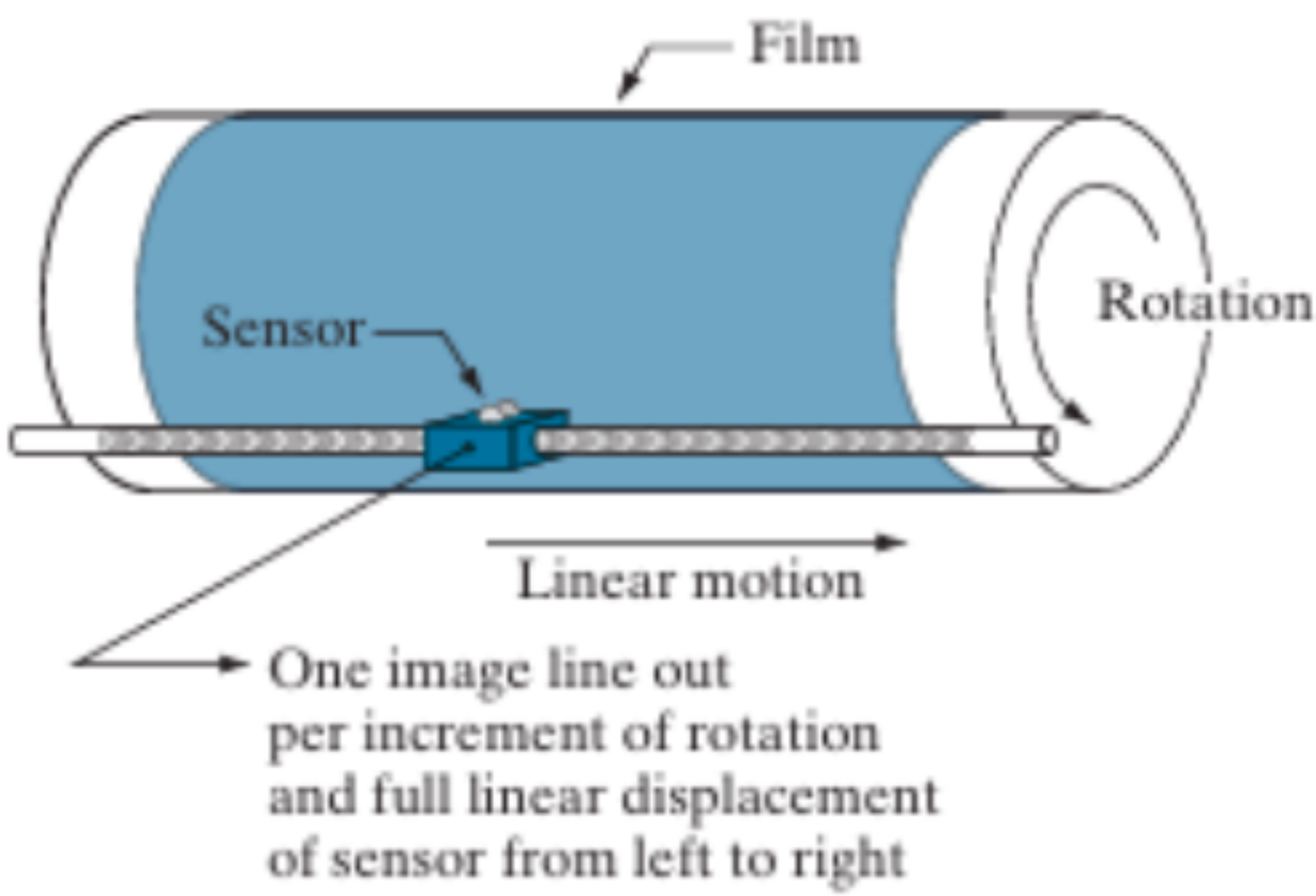
Define image sensor

An image sensor converts light or electromagnetic energy into electrical energy, often relying on photo-converter materials for visible light.

Single Sensor Acquisition



Principle: It uses **one sensor** and **mechanical movement** to scan the image.

Process:

- **Rotation:** The drum rotates to capture one line.

- **Linear Motion:** The sensor moves sideways to the next line.

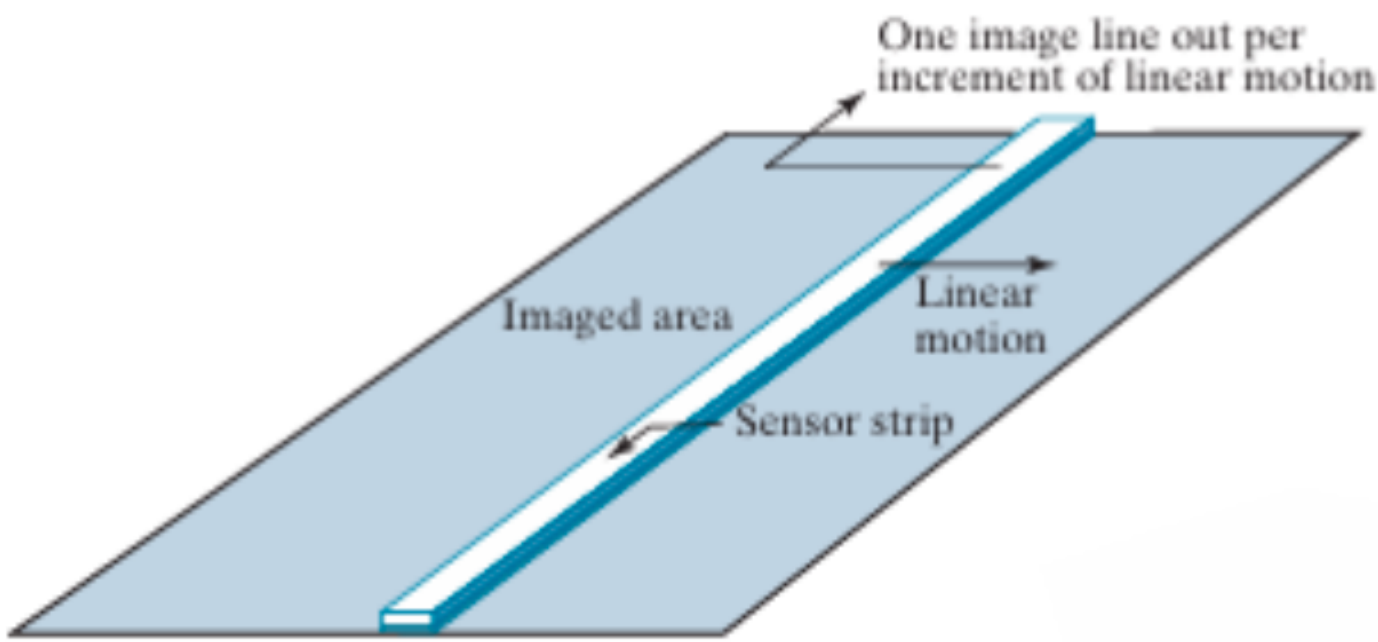
Result: One full rotation = **One image line**.

Pros: **Inexpensive** (Low cost).

Cons: **Slow** (Takes time to move).

Example: Drum Scanner.

linear sensor strip



Mechanism: Uses a **strip (row)** of many sensors.

Motion: Only **one** linear motion is needed.

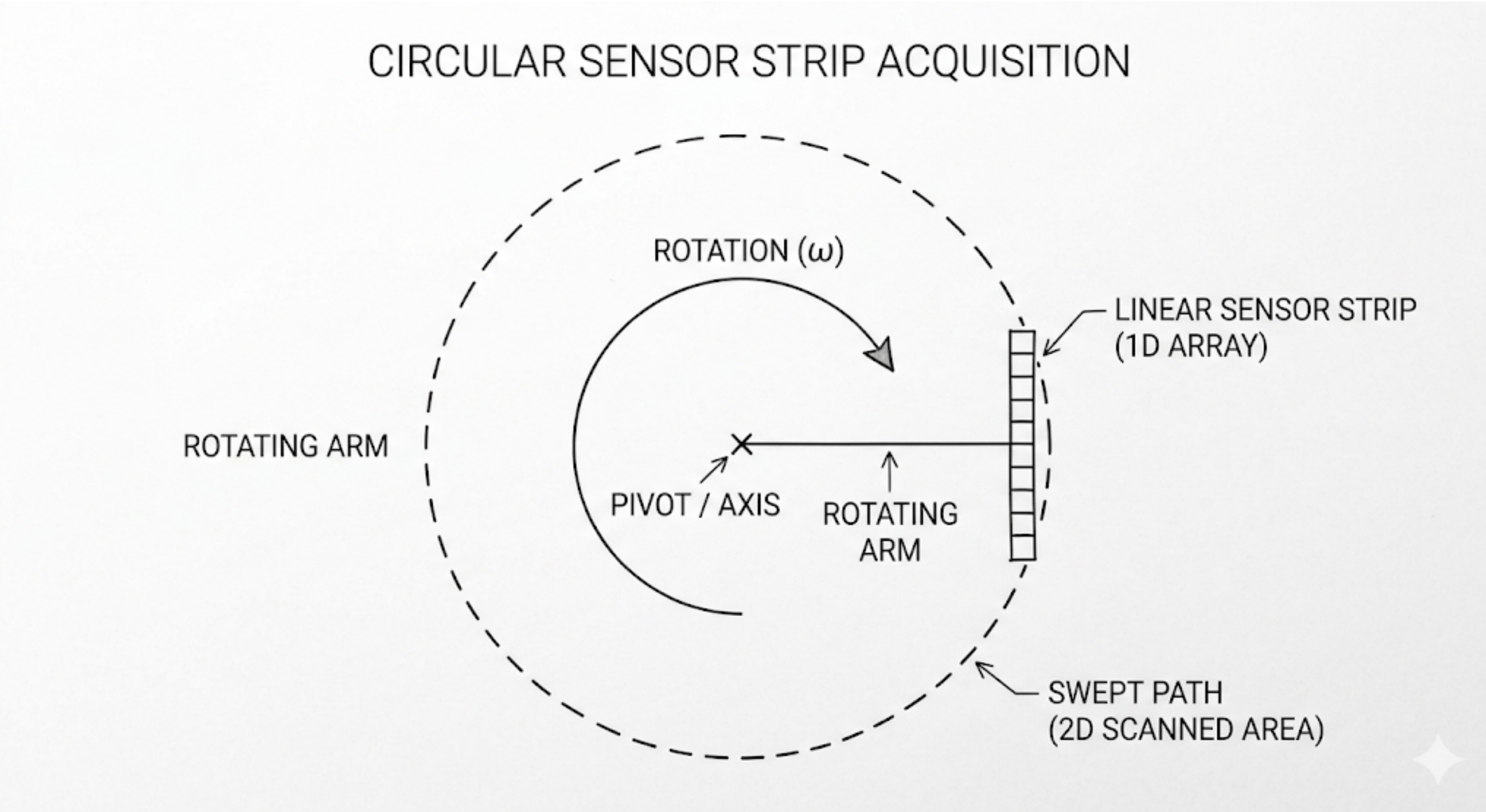
Process: Scans an **entire row** at once.

Pros: Much **faster** than single sensor.

Cons: More **expensive** (more sensors needed).

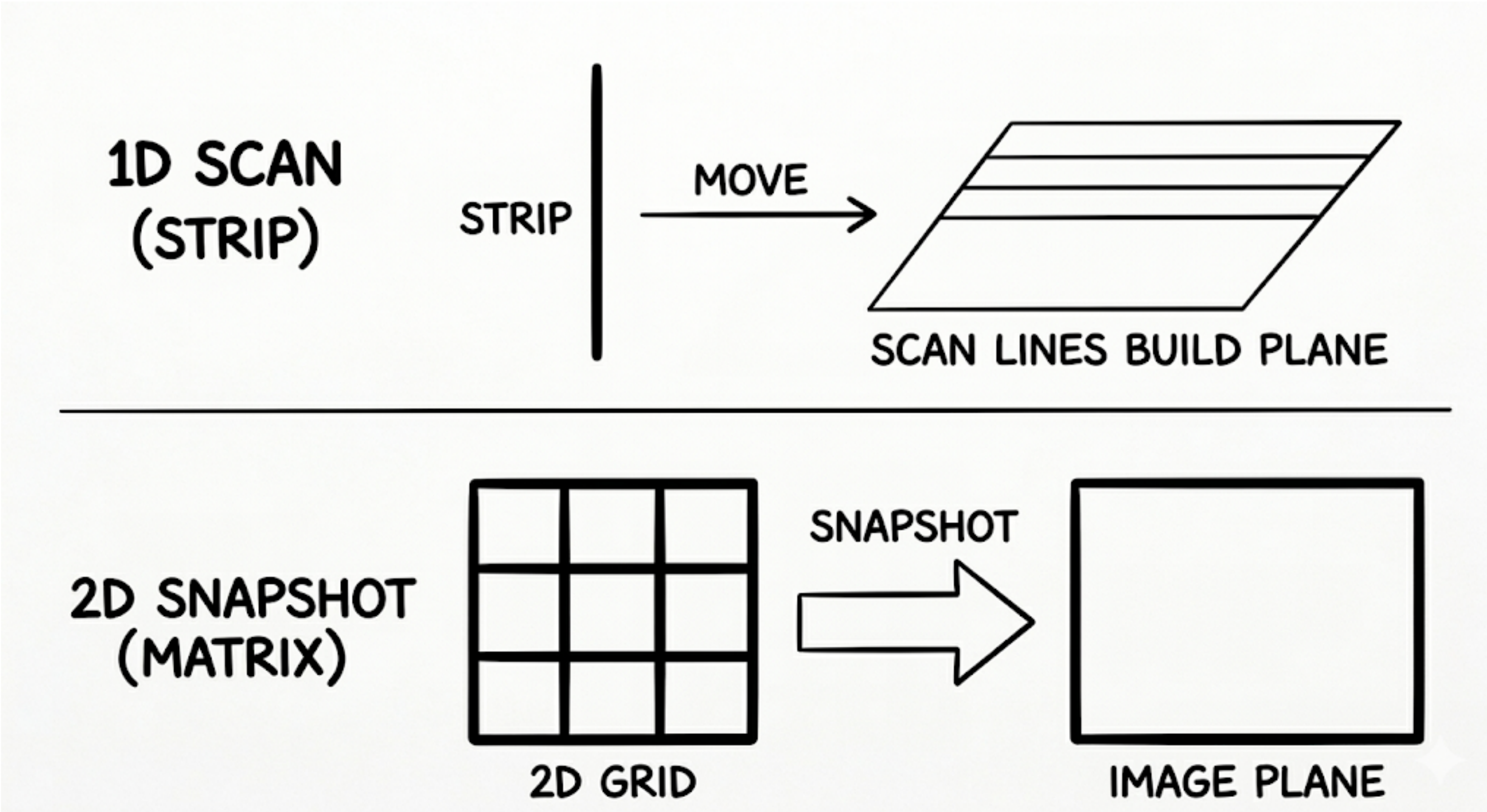
Example: Flatbed Scanner.

Circular Sensor Strip Acquisition



Mechanism: Sensors are arranged in a **circular/ring** shape.
Motion: Only **rotational motion** is needed.
Process: Captures an entire **cross-section (slice)** of the object at once.
Pros: Extremely **fast** for 3D reconstruction.
Cons: Very **expensive** and complex hardware.
Example: Medical **CT (Computed Tomography) Scanners**.

Sensor Arrays



Mechanism: Sensors are arranged in a **2D Grid (Matrix)**.
Motion: **No mechanical motion** is required to capture an image.
Process: Captures the **entire scene instantly** (all pixels at once).
Pros: **Extremely fast** (Instantaneous capture).
Cons: Most **expensive** and complex to manufacture.
Example: **Digital Cameras**, Smartphones.